

Amit Kumar

iamitkumar2007@gmail.com | +919889460738 | linkedIn | Github

Education

Maulana Azad National Institute of Technology , Bhopal, Master of Technology in Artificial Intelligence	Aug 2023 – July 2025
Institute of Engineering and Technology, Bundelkhand University, Jhansi, Bachelor of Technology in Computer Science and Engineering	Aug 2018 – July 2022

Internship Experience

Full Stack Developer Intern
[Wabtec Corporation], Onsite
Sep 2024 – Aug 2025

Projects

NL2SQL Chatbot	July 2025
- Developed a LLM-powered SQL Query Generator using Python, Streamlit, and SQLite. - Converted 100+ natural language prompts into optimized SQL queries with real-time tabular visualization. - Implemented memory-efficient prompt handling to reduce latency and improve system responsiveness. - Designed validation pipelines to catch malformed queries and mitigate SQL injection & adversarial attacks. - Achieved 80% improvement in query accuracy and 60% reduction in execution errors through prompt engineering and validation.	
Cold Email Generator	Jun 2025
- Built an AI-powered Streamlit web application that generates personalized cold emails from job descriptions using Groq's LLaMA-3 (70B). - Implemented job data extraction via LangChain and real-time portfolio querying using ChromaDB vector database. - Used NLP preprocessing and chunked inference to process long career pages, ensured robustness with exception handling. - Integrated a tech stack including LangChain, Groq API, and Streamlit for a complete LLM-powered automation tool.	
Movie Mentor	Feb 2024
- Developed a content-based Movie Recommender System using Python, Pandas, and Scikit-learn. - Preprocessed and merged movie metadata for comprehensive feature extraction. - Utilized Count Vectorizer and cosine similarity to compute movie recommendations. - Deployed the recommendation system using Streamlit for real-time user interaction.	
Asset Management System	Feb 2025
- Built a secure Asset Management Tool for Wabtec Corporation using React.js, PostgreSQL, and Docker. - Integrated role-based access control with admin privileges and Okta-based authentication. - Integrated SSL certificate for secure communication and automated email alerts on low asset quantity. - Utilized GitLab for version control and CI/CD, ensuring efficient development and deployment.	

Technical Skills

- Languages:** Python, C++, JavaScript
- Deep Learning / Machine Learning Techniques:** CNN, RNN, NLP, Transfer Learning, LLMs
- Deep Learning / Machine Learning Libraries:** Scikit-learn, TensorFlow, Pandas, NumPy, Matplotlib
- Frameworks / Tools:** ReactJS, LangChain, Streamlit, FastAPI, LangGraph
- DevOps:** GitLab, Docker, Github
- Database :** PostgreSQL, DBMS concepts, Chroma (Vector DB), FAISS (Vector DB)
- Cloud:** AWS (S3, Lambda, EC2, Bedrock, Vector DB)